

SQL Shell (psql)

Server [localhost]:
Database [postgres]:
Port [5432]:
Username [postgres]:
Password for user postgres:

psql (18.0)
WARNING: Console code page (437) differs from Windows code page (1252)
8-bit characters might not work correctly. See psql reference
page "Notes for Windows users" for details.
Type "help" for help.

postgres=# CREATE DATABASE ecommerce_db;
CREATE DATABASE
postgres=# \c ecommerce_db
You are now connected to database "ecommerce_db" as user "postgres".
ecommerce_db=# CREATE TABLE online_retail (invoice TEXT,stock_code TEXT,description TEXT,quantity INT,invoice_date TIMESTAMP,unit_price
NUMERIC(10,2),customer_id TEXT,country TEXT);
CREATE TABLE
ecommerce_db=# \copy online_retail(invoice, stock_code, description, quantity, invoice_date, unit_price, customer_id, country)FROM 'C:/Users/reshw/Documents
/sql_data/online_retail_II.csv'CSV HEADER;
COPY 1067371
ecommerce_db=# SELECT * FROM online_retail LIMIT 5;

invoice	stock_code	description	quantity	invoice_date	unit_price	customer_id	country
489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

(5 rows)

ecommerce_db=# SELECT COUNT(*) FROM online_retail;
count

1067371
(1 row)

ecommerce_db=# SELECT invoice, customer_id, country FROM online_retail WHERE country = 'United Kingdom' ORDER BY invoice_date DESC LIMIT 10;

invoice	customer_id	country
581586	13113.0	United Kingdom
581586	13113.0	United Kingdom
581586	13113.0	United Kingdom
581586	13113.0	United Kingdom
581585	15804.0	United Kingdom
581585	15804.0	United Kingdom
581585	15804.0	United Kingdom
581585	15804.0	United Kingdom
581585	15804.0	United Kingdom
581585	15804.0	United Kingdom

(10 rows)

ecommerce_db=# SELECT country, SUM(quantity * unit_price) AS revenue FROM online_retail GROUP BY country ORDER BY revenue DESC LIMIT 10;

country	revenue
United Kingdom	16382583.88
EIRE	615519.55
Netherlands	548524.95
Germany	417988.56
France	328191.80
Australia	167129.07
Switzerland	99728.76
Spain	91859.48
Sweden	87809.42
Denmark	65741.09

(10 rows)

```
ecommerce_db=# CREATE TEMP TABLE products AS SELECT DISTINCT stock_code, description FROM online_retail;
SELECT 9021
```

```
ecommerce_db=# SELECT r.invoice, p.description, r.quantity, r.unit_price FROM online_retail r INNER JOIN products p ON r.stock_code = p.stock_code LIMIT 10;
```

invoice	description	quantity	unit_price
489434	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	6.95
489434	Unsaleable, destroyed.	12	6.75
489434	PINK CHERRY LIGHTS	12	6.75
489434		12	6.75
489434		12	6.75
489434	WHITE CHERRY LIGHTS	12	6.75
489434	Unsaleable, destroyed.	12	6.75
489434	WHITE CHERRY LIGHTS	12	6.75
489434	RECORD FRAME 7" SINGLE SIZE	48	2.10
489434	STRAWBERRY CERAMIC TRINKET POT	24	1.25

(10 rows)

```
ecommerce_db=# SELECT r.invoice, r.stock_code, p.descriptionFROM online_retail r LEFT JOIN products p ON r.stock_code = p.stock_code LIMIT 10;
ERROR:  syntax error at or near "r"
LINE 1: ...ce, r.stock_code, p.descriptionFROM online_retail r LEFT JOI...
                                     ^
```

```
ecommerce_db=# SELECT p.stock_code, p.description, r.invoice FROM online_retail r RIGHT JOIN products p ON r.stock_code = p.stock_code LIMIT 10;
```

stock_code	description	invoice
ALL 20 LIGHTS	489434	85048 15CM CHRISTMAS GLASS B
79323P	Unsaleable, destroyed.	489434
79323P	PINK CHERRY LIGHTS	489434
79323P		489434
79323W		489434
79323W	WHITE CHERRY LIGHTS	489434
79323W	Unsaleable, destroyed.	489434
79323W	WHITE CHERRY LIGHTS	489434
22041	RECORD FRAME 7" SINGLE SIZE	489434
21232	STRAWBERRY CERAMIC TRINKET POT	489434

(10 rows)

```
ecommerce_db=# SELECT customer_id, total_spent FROM ( SELECT customer_id, SUM(quantity * unit_price) AS total_spent FROM online_retail WHERE customer_id IS NOT NULL GROUP BY customer_id ) AS sub WHERE total_spent > 5000 ORDER BY total_spent DESC;
```

customer_id	total_spent
18102.0	598215.22
14646.0	523342.07
14156.0	296564.69
14911.0	270248.53
17450.0	233579.39
13694.0	190825.52
17511.0	171885.98
12415.0	143269.29
16684.0	141502.25
15061.0	136391.48
15311.0	113513.07
13089.0	113214.19
17949.0	98895.59
16029.0	91800.91
14298.0	90489.31
15769.0	84269.38
13798.0	73573.47
15838.0	73404.11
12931.0	71299.67
17841.0	69516.19
14088.0	62400.02
16422.0	61110.54
13081.0	58328.15
14096.0	57120.91
16754.0	56560.58
13777.0	56016.57
17850.0	55703.13
14680.0	54801.34
17389.0	54587.03
13093.0	54073.73
16013.0	52788.92
13408.0	50852.85
12748.0	49970.13
17404.0	46006.82
13881.0	44392.50

```
^Ccancel request sent
ecommerce_db=# SELECT AVG(order_total) AS avg_order_value FROM ( SELECT invoice, SUM(quantity * unit_price) AS order_total FROM online_retail GROUP BY invoice ) AS orders;
      avg_order_value
-----
 359.6488877079137764
(1 row)
```

```
ecommerce_db=# SELECT description, SUM(quantity) AS total_quantity FROM online_retail GROUP BY description ORDER BY total_quantity DESC LIMIT 10;
      description      | total_quantity
-----+-----
WORLD WAR 2 GLIDERS ASSTD DESIGNS |      108545
WHITE HANGING HEART T-LIGHT HOLDER |       93050
ASSORTED COLOUR BIRD ORNAMENT      |       81306
JUMBO BAG RED RETROSPOT            |       78090
BROCADE RING PURSE                 |       70700
PACK OF 60 PINK PAISLEY CAKE CASES |       56575
60 TEATIME FAIRY CAKE CASES        |       54366
SMALL POPCORN HOLDER               |       49616
PACK OF 72 RETROSPOT CAKE CASES    |       49344
PACK OF 72 RETRO SPOT CAKE CASES   |       46106
(10 rows)
```


WORLD WAR 2 GLIDERS ASSTD DESIGNS	108545
WHITE HANGING HEART T-LIGHT HOLDER	93050
ASSORTED COLOUR BIRD ORNAMENT	81306
JUMBO BAG RED RETROSPOT	78090
BROCADE RING PURSE	70700
PACK OF 60 PINK PAISLEY CAKE CASES	56575
60 TEATIME FAIRY CAKE CASES	54366
SMALL POPCORN HOLDER	49616
PACK OF 72 RETROSPOT CAKE CASES	49344
PACK OF 72 RETRO SPOT CAKE CASES	46106

(10 rows)

```
ecommerce_db=# CREATE OR REPLACE VIEW monthly_revenue AS SELECT DATE_TRUNC('month', invoice_date) AS month,SUM(quantity * unit_price) AS revenue FROM online_retail GROUP BY month ORDER BY month;
CREATE VIEW
ecommerce_db=# SELECT * FROM monthly_revenue LIMIT 12;
```

month	revenue
2009-12-01 00:00:00	799847.11
2010-01-01 00:00:00	624032.89
2010-02-01 00:00:00	533091.42
2010-03-01 00:00:00	765848.76
2010-04-01 00:00:00	590580.43
2010-05-01 00:00:00	615322.83
2010-06-01 00:00:00	679786.61
2010-07-01 00:00:00	575236.36
2010-08-01 00:00:00	656776.34
2010-09-01 00:00:00	853650.43
2010-10-01 00:00:00	1045168.35
2010-11-01 00:00:00	1422654.64

(12 rows)

```
ecommerce_db=# CREATE INDEX idx_customer_id ON online_retail(customer_id);
CREATE INDEX
ecommerce_db=# CREATE INDEX idx_invoice_date ON online_retail(invoice_date);
CREATE INDEX
ecommerce_db=#
```